

Exponential Functions and Their Graphs

ANSWER KEY



Evaluating and describing

- 1 Let $f(x) = 3 \cdot 2^x$. Evaluate:
- a** $f(0) = 3$ **b** $f(3) = 24$ **c** $f(-2) = \frac{3}{4}$
- 2 For $y = 5^x$, state:
- a** the domain All real numbers. **b** the range $y > 0$.
- c** the horizontal asymptote $y = 0$. **d** the y -intercept $(0, 1)$.
- 3 Describe how the graph of $g(x) = 2^{x-3} + 1$ is obtained from the graph of $y = 2^x$, and state the asymptote of g .
- Shift right 3 units and up 1 unit. Horizontal asymptote: $y = 1$.
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Identifying from graphs and applications

- 4 The graph below passes through $(0, 1)$, $(1, 2)$, and $(2, 4)$. Write the equation of the exponential function shown.
- $y = 2^x$.
- 5 An investment of 1000 riyals grows with continuous compounding at an annual rate of 6%:
- $A(t) = 1000e^{0.06t}$. Find the value after 5 years, to the nearest rial.
- $A(5) = 1000e^{0.3} \approx 1350$ riyals.
- 6 Determine whether each function represents growth or decay:
- a** $y = 4(1.08)^x$ Growth. **b** $y = 7(0.85)^x$ Decay.
- c** $y = 2e^{-0.3x}$ Decay. **d** $y = \left(\frac{5}{4}\right)^x$ Growth.
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Compound interest

- 7 2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P\left(1 + \frac{r}{n}\right)^{nt}$.
- a** Find the value after 3 years, to the nearest rial. $A = 2000\left(1 + \frac{0.05}{12}\right)^{36} \approx 2000(1.1616) \approx 2323$ riyals.
- b** Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest rial.
- $A = 2000e^{0.15} \approx 2000(1.1618) \approx 2324$ riyals.

- 8 How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.
- $$P = \frac{10000}{(1.04)^8} \approx \frac{10000}{1.3686} \approx 7308 \text{ riyals.}$$
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Comparing exponential growth rates

- 9 Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.
- 4%: $500(1.04)^{10} \approx 740$ riyals. 7%: $500(1.07)^{10} \approx 984$ riyals. Difference ≈ 244 riyals.
- 10 Rank the following functions from slowest to fastest growth as $x \rightarrow \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.
- Any exponential with base > 1 eventually outgrows any polynomial, no matter how slowly it starts. So even though 1.01^x grows very slowly at first, it eventually overtakes x^{10} . Long-run ranking (slowest to fastest): $x^{10} < 1.01^x < 2^x$.